

Q1. A line passes through the point $A(2, -1, 5)$ and is parallel to the vector $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j} + \mathbf{k}$. Express the equation of this line in Cartesian form.

Q2. Find the angle between the vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$.

Q3. A plane is defined by the Cartesian equation $4x - 7y + 2z = 15$. Find a vector equation for this plane in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$.

Q4. Convert the line given by the Cartesian equation $\frac{x+3}{2} = \frac{5-y}{3} = z + 1$ into its equivalent vector form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$.

Q5. A plane passes through the point $(1, 0, -2)$ and has a normal vector $\mathbf{n} = \begin{pmatrix} 6 \\ -1 \\ 3 \end{pmatrix}$.

(a) What is the equation of the plane in scalar product form $\mathbf{r} \cdot \mathbf{n} = d$?

(b) Express the equation of the plane in Cartesian form.

Q6. The vectors $\mathbf{p} = 4\mathbf{i} + k\mathbf{j} - 2\mathbf{k}$ and $\mathbf{q} = k\mathbf{i} - 5\mathbf{j} + 3\mathbf{k}$ are perpendicular. Determine the value of the constant k .

Q7. The line L_1 passes through the points $A(0, 5, 2)$ and $B(4, -3, 0)$. The line L_2 is defined by the Cartesian equation $\frac{x-1}{2} = \frac{y+2}{1} = \frac{z-6}{-2}$. If L_1 and L_2 intersect, state the coordinates of their point of intersection. Else, deduce whether the lines are parallel or skew.

Q8. Consider the lines:

$$\mathbf{r}_1 = (2\mathbf{i} + \mathbf{j} - \mathbf{k}) + s(\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$$

$$\mathbf{r}_2 = (5\mathbf{i} + \mathbf{j} + \alpha\mathbf{k}) + t(2\mathbf{i} - \mathbf{j} + \mathbf{k})$$

Find the value of the constant α such that the two lines intersect.

Q9. The vector equation of a line is $\mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ -4 \end{pmatrix} + t \begin{pmatrix} 1 \\ -3 \\ 0 \end{pmatrix}$. Express the equation of this line in Cartesian form.

Q10. Given that $|\mathbf{a}| = 5$, $|\mathbf{b}| = 8$, and the angle between $\mathbf{a}$ and $\mathbf{b}$ is 120° , calculate the exact value of:

$$(\mathbf{a} + 2\mathbf{b}) \cdot (3\mathbf{a} - \mathbf{b})$$